Variety $\longleftrightarrow$ Ideal

$X \subseteq k^n \longmapsto I(X) = \{ f \in k[x_1, \dots, x_n] : f \text{ vanishes on } X \}$

$V(f_1, \dots, f_m) = \{ a \in k^n : f_1(a) = \dots = f_m(a) = 0 \}$

$\langle f_1, \dots, f_m \rangle \longmapsto V(f_1, \dots, f_m) \subseteq k^n \longmapsto I(V(f_1, \dots, f_m))$

the set of polynomials that vanish whenever the polynomials $f_1, \dots, f_m$ vanish.

One would frequently study a variety $V \subseteq k^n$ by studying the respective ideal $I(V)$.

1.4.10 Lemma For $f_1, \dots, f_s \in k[x_1, \dots, x_n]$, one has

$$\langle f_1, \dots, f_s \rangle \subseteq I(V(f_1, \dots, f_s)).$$

Moreover, there exist choices of $f_1, \dots, f_s$ where this inclusion is strict.

Proof: $g \in \langle f_1, \dots, f_s \rangle \iff g = f_1 h_1 + \dots + f_s h_s$ for some $h_1, \dots, h_s \in k[x_1, \dots, x_n]$.

$g \in I(V(f_1, \dots, f_s)) \iff g$ vanishes on $V(f_1, \dots, f_s)$

$\iff g(a) = 0$ whenever $f_1(a) = \dots = f_s(a) = 0$, for $a \in k^n$.

This explains the inclusion in the assertion.

For strict inclusion, consider $k[x]$ with $V(x^2)$:

$\langle x^2 \rangle$ = set of all polynomials that do not have constant and linear terms.

$V(x^2) = \{0\} \Rightarrow I(V(x^2)) =$ set of polynomials

that do not have the constant term. Consequently,

$x \in I(V(x^2))$ but $x \notin \langle x^2 \rangle$. □

The so called Nullstellensatz clarifies the relation of $I(V(f_1, \dots, f_s))$ to $\langle f_1, \dots, f_s \rangle$ in the algebraic terms in the case when the field k is algebraically closed (for example, for $k = \mathbb{C}$).

1.4.11 Proposition Let $V, W \subseteq k^n$ be affine varieties. Then the following hold:

(i) $V \subseteq W \Rightarrow I(V) \supseteq I(W)$.

(ii) $V = W \Leftrightarrow I(V) = I(W)$.

Proof: (i) follows directly from our definition of $I(X)$ for a set $X \subseteq k^n$.

(ii) $V = W \Rightarrow I(V) = I(W)$ is trivial.

Conversely, assume $I(V) = I(W)$. Let $V = V(f_1, \dots, f_s)$ and $W = V(g_1, \dots, g_t)$ with $f_1, \dots, f_s, g_1, \dots, g_t \in k[x_1, \dots, x_n]$.

Equality $I(V) = I(W)$ means that every polynomial vanishing on V also vanishes on W and vice versa

$f_1, \dots, f_s$ vanish on $V \Rightarrow$ (by $I(V) = I(W)$) $f_1, \dots, f_s$ vanish on W. So $W \subseteq V(f_1, \dots, f_s) = V$.

Exchanging the roles of V and W, we also have the other inclusion $V \subseteq W$.

$\Rightarrow V = W$. □

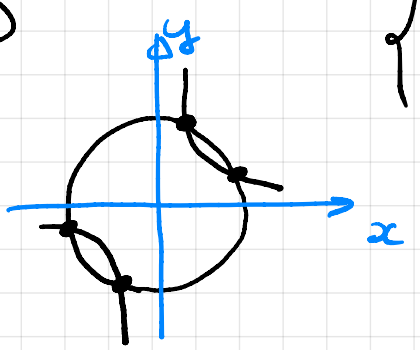
$f_1 = \dots = f_s = 0$ $\longrightarrow$ $\langle f_1, \dots, f_s \rangle$

$\downarrow$ $\cap$

$V(f_1, \dots, f_s)$ $\longrightarrow$ $I(V(f_1, \dots, f_s))$

Nullstellensatz (for k algebraically closed).

1.4.12 Exercises

① 

$$\begin{cases} x^2+y^2-1 = 0 \\ 2xy - 1 = 0 \end{cases}$$

Find a polynomial $f(x) \in k[x] \setminus \{0\}$ such that the equation $f(x) = 0$ follows from this system.

Is $f \in \langle x^2+y^2-1, 2xy-1 \rangle$?

② $k = \mathbb{F}_2$, k^2

$I(k^2) = ???$ Describe $I(k^2)$ by a set of generators.

1.5 Polynomials in one variable

1.5.1. Def For $f = a_0 + a_1 x + \dots + a_m x^m \in k[x] \setminus \{0\}$ with $a_0, \dots, a_m \in k$ and $a_m \neq 0$, we call $a_m x^m$ the leading term of f. Notation: $LT(f)$.

1.5.2 Prop For every choice of $f \in k[x]$ and $g \in k[x] \setminus \{0\}$ there exists a unique decomposition

$$f = g \cdot q + r \quad \text{with } q, r \in k[x]$$

and $\deg r < \deg g$. Furthermore, there exists

an algorithm that calculates q and r for given f and g.

Proof: In our algorithm we start with

$q = 0$ and $r = f$

q: quotient in progress

r: remainder in progress.

We will maintain the equality $f = g \cdot q + r$ as the invariant, making $\deg r$ smaller in each iteration until the condition $\deg r < \deg g$ is satisfied.

If $\deg r < \deg g$, then we can terminate.

Otherwise, we want to update r using g to make $\deg r$ smaller.

$$\deg r \geq \deg g \quad \Rightarrow$$

$$f = g \cdot \underbrace{\left(q + \frac{LT(r)}{LT(g)}\right)}_{\text{new } q} + \underbrace{\left(r - \frac{LT(r)}{LT(g)} g\right)}_{\text{new } r}$$

(by taking into account $f = g \cdot q + r$).

This leads to:

Univariate_Division (f, g):

Assumptions: $f, g \in k[x]$, $g \neq 0$

Result: quotient q and the remainder r of the long division of f by g.

1: $r := f$ ▷ Remainder in progress

2: $q := 0$ ▷ Quotient in progress

3: ▷ Cycle invariant: $f = g \cdot q + r$

4: while $\deg r \geq \deg g$

5: $T := \frac{LT(r)}{LT(g)}$

6: $q := q + T$ ▷ Update of the quotient is progress

7: $r := r - T \cdot g$ ▷ Update of the remainder is progress

8: end while

9: return q, r

The algorithm is correct and is terminating in finitely many steps, because we maintain the invariant $f = g \cdot q + r$ and the updates of q and r are such that $\deg r$ gets strictly smaller with each iteration.

It remains to show the uniqueness of the decomposition.

Assume $f = g \cdot q + r = g \cdot Q + R$

with $q, r, Q, R \in k[X]$ and $\deg r < \deg g$, $\deg R < \deg g$.

If $q = Q$, then $r = f - g \cdot q = f - gQ = R$

and so $r = R$.

Assume $q \neq Q$. Then

$$0 = f - f = (g \cdot Q - R) - (g \cdot q - r)$$
$$= g \cdot (Q - q) + r - R$$

$$\Rightarrow \underbrace{R - r}_{\text{has deg} < \deg g} = \underbrace{g \cdot (Q - q)}_{\neq 0}.$$

$\neq 0$, has $\deg g$

The left-hand side has degree $< \deg g$, but the right-hand side has degree $\geq \deg g$.

$\Rightarrow$ Contradiction.

$\Rightarrow Q = q$. □

1.5.3. Cor Every $f \in k[x] \setminus \{0\}$ has at most $\deg f$ roots in k.

Proof: Induction on $m := \deg f$. For $m = 0$, this is clear. Let $m \geq 1$ and assume that the assertion is verified for polynomials of degree $< m$.

Consider any $f \in k[x] \setminus \{0\}$ of $\deg f = m$.

If f has no roots in k, we are done, because $0 \leq m$.

If f has at least one root $a \in k$, we can use long division to divide f by $x - a$.

We obtain $f = (x-a) \cdot q + r$ where $q \in k[x]$ and $\deg r < \deg(x-a) = 1$, meaning $r \in k$.

Since $f(a) = 0$, we have $f(a) = (a-a) \cdot q(a) + r$

$\Rightarrow r = 0$.

$\Rightarrow f = (x-a) \cdot q$, where q is a polynomial of degree $m-1$. By induction, q has at most $m-1$ roots in k

$\Rightarrow f$ has at most m roots (the root a and the roots of q). $\square$

1.5.4. Corollary Every ideal I in $k[x]$ has the form $I = \langle h \rangle$ for some $h \in k[x]$.

Proof: If $I = \{0\}$, the assertion is trivial as we can fix $h = 0$. Assume $I \neq \{0\}$. Choose a polynomial $h \in I$ with $h \neq 0$ with the lowest possible degree.

We claim that $I = \langle h \rangle$.

The inclusion $I \supseteq \langle h \rangle$ is clear because $h \in I$.

Let's show the converse: $I \subseteq \langle h \rangle$.

Consider any $f \in I$. We can use long division to divide f by h: $f = h \cdot q + r$ with $q, r \in k[x]$ and $\deg r < \deg h$. But then

$$r = \underbrace{f}_{\in I} - \underbrace{h \cdot q}_{\in I} \in I$$ has smaller degree than h,

contradiction to the choice of h whenever $r \neq 0$. So, $r = 0$.

$$\Rightarrow f = h \cdot q + 0 = h \cdot q \in \langle h \rangle. \qquad \square$$

1.5.5. Def. A principal ideal of a ring is an ideal that can be generated by just one element of the ring. A principal ideal domain (PID) is a commutative ring, in which every ideal is principal.

We have shown: $k[x]$ is a PID.

Another example of a PID is $\mathbb{Z}$.

1.5.6. Def Greatest common divisor (abbr.: gcd) of $f, g \in k[x]$ is a polynomial $h \in k[x]$ such that:

(i) h divides both f and g

(ii) every $p \in k[x]$ that divides f and g, also divides h.

Notation: $h = \gcd(f, g)$

The notation suggests that $\gcd(f,g)$ is unique, but we will see that $\gcd(f,g)$ is unique up to constant non zero multiple only. Thus, by writing $\gcd(f,g)$ we abuse notation (but we won't misuse it hopefully).

In math: abuse of notation – tentatively slightly wrong notation in correct arguments

misuse of notation – wrong notation that leads to an incorrect argument.

The next proposition is a univariate prototype of what we are going to develop in the case of n variables. The proposition explains gcd in terms of ideals and it addresses the question on how to do computations with ideals in the ring $k[x]$.

1.5.7 Proposition Let $f, g \in k[x]$. Then the following hold:

(i) $\gcd(f,g)$ exists and is unique up to a non-zero constant multiple (i.e. up to a factor in $k \setminus \{0\}$).

(ii) $\gcd(f,g)$ generates the ideal $\langle f,g \rangle$, meaning that $\langle \gcd(f,g) \rangle = \langle f,g \rangle$ (this is an equivalent description of the gcd in the language of ideals).

(iii) There exists an algorithm that determines $\gcd(f,g)$ for given f and g.

Proof: Note that f and g may both be 0. We have not excluded this possibility in our assumptions. This is a boundary case we deal with separately.

Note that f being divisible by h means that $f = h \cdot q$ for some $q \in k[x]$. This also applies to the degenerate situations. For example, $0 = 0 \cdot 1$ means that 0 is divisible by 0. If we keep this in mind, we see that $\gcd(0,0) = 0$ is 0 is divisible by 0 and every polynomial divides 0. On the other hand,

if f or g is not 0, then $\gcd(f,g) \neq 0$. Say, if $f \neq 0$, then f is not divisible by zero, as $f \neq q \cdot 0$ for every $q \in k[x]$, which implies that 0 is not a common divisor of f and g. The above shows that

$$\gcd(f,g) = 0 \iff f = g = 0.$$

In this degenerate case all three assertions (i), (ii) and (iii) are true. So, in what follows we may assume $\gcd(f,g) \neq 0$. The plan for the proof is to fix a polynomial h satisfying $\langle h \rangle = \langle f, g \rangle$ and then to verify that this polynomial is the $\gcd(f,g)$ obtaining (ii). Once this is settled we will show (i) and eventually derive an algorithm for computing $\gcd(f,g)$ to establish (iii).

Corollary 1.5.4 tells us that every ideal $I \subseteq k[x]$ can be generated by just one polynomial. Applying this to $I = \langle f, g \rangle$ we see that $\langle h \rangle = \langle f, g \rangle$ for some $h \in k[x]$ without yet knowing how h is related to f and g in a more elementary fashion (via gcd) and without knowing how to construct h algorithmically. Let's see why h is $\gcd(f,g)$. Before we see that h is the greatest common divisor of f and g, we first check that it is a common divisor.

(ii): The set equality $\langle h \rangle = \langle f, g \rangle$ is two inclusions $\langle h \rangle \subseteq \langle f, g \rangle$ and $\langle h \rangle \supseteq \langle f, g \rangle$. The inclusion $\langle h \rangle \supseteq \langle f, g \rangle$ implies that $f \in \langle h \rangle$ and $g \in \langle h \rangle$. Hence, h is a multiple of f and a multiple of g. Thus, h is indeed a common divisor of f and g. Now, consider any common divisor p of f and g. We have $f = Cp$, $g = Dp$ for some $C, D \in K[x]$. Inclusion $\langle h \rangle \subseteq \langle f, g \rangle$ tells us that $h \in \langle f, g \rangle$, meaning that $h = Af + Bg$ for some $A, B \in K[x]$. Putting things together we have:

$$h = Af + Bg = ACp + BDp = (AC + BD) \cdot p.$$

Thus, h is divisible by p. It follows that h is the greatest common divisor of f and g.

(i): Why uniqueness up to a non zero constant multiple? Remember that we assume $\gcd(f, g) \neq 0$ as the case $\gcd(f, g) = 0$ is already done. If h and H are both greatest common divisors then h divides H and H divides h. As $\gcd(f, g) \neq 0$, both are non zero polynomials. So, h dividing H implies $\deg h \leq \deg H$ and H dividing h implies $\deg H \leq \deg h$. So, $\deg h = \deg H$. H being divisible by h implies $H = h \cdot q$ for some $q \in K[x]$. But then $\deg H = \deg(h \cdot q) = \deg h + \deg q$

In view of $\deg 1 = \deg h$, we obtain $\deg q = 0$, meaning that $q \in k \setminus \{0\}$. This concludes the proof of (ii).

(iii) We are going to describe the standard algorithm for computing $\gcd(f,g)$, the so called **Euclidean Algorithm**. It is an iterative algorithm that uses long division in each iteration.

$\gcd(f,0) = f$, so when $g = 0$, we are done. If however $g \neq 0$, we can divide f by g obtaining the quotient $q \in k[x]$ and the remainder $r \in k[x]$ satisfying

$$f = q \cdot g + r, \quad \deg r < \deg g.$$

Using q, g, r our $\gcd(f,g)$ is now expressed as $\gcd(f,g) = \gcd(q \cdot g + r, g)$.

We claim that $\gcd(q \cdot g + r, g) = \gcd(r, g)$.

Why is this true? Every polynomial p that divides r and g also divides $q \cdot g + r$ and g. Conversely, every polynomial p that divides $q \cdot g + r$ and g also divides

$r = (q \cdot g + r) - q \cdot g$ and g. It follows that the pairs $(q \cdot g + r, g)$ and (r, g) have the same set of common divisors. Thus, they also have the same greatest

common divisor. We have

$\gcd(f,g) = \gcd(g,r)$. This observation can be used to set up an iterative procedure: g is going to become the new f and r the new g. During the iteration, the pair of the polynomials being processed will change, but the gcd of the pair will stay the same, while the degree of the second polynomial of the pair will drop with each iteration (because $\deg g > \deg r$) until the case $g = 0$ is reached. Here is the complete algorithm:

Greatest Common Divisor (f, g):

Assumption: $f, g \in k[x]$

Result: $\gcd(f, g)$.

1: ▷ Invariant of the while loop:
$\gcd(f, g)$ does not change.

2: while $g \neq 0$:

3: $q, r := \text{Long Division}(f, g)$

4: $f := g$

5: $g := r$

6: ▷ $\deg g$ has dropped:
we are making progress;
see you in the next iteration ☺

7: end while

8: return f ▷ $\gcd(f, g) = f$ when $g = 0$.

There is not much to be said on why this algorithm correctly determines $\gcd(f,g)$ after finitely many iterations. The discussion before the algorithm and the comment lines explain it all. □

The case of an ideal $\langle f,g\rangle$ generated by two polynomials f and g in $k[x]$ is the base case for the case of an ideal $\langle f_1,\dots,f_s\rangle$ generated by finitely many polynomials $f_1,\dots,f_s \in k[x]$. In this more general situation gcd is still the concept that "explains everything".

1.5.8 Def Greatest common divisor of finitely many polynomials $f_1,\dots,f_s \in k[x]$ is a polynomial $h \in k[x]$ such that:

(i) h divides all of the $f_1,\dots,f_s$ (it is a common divisor)

(ii) Every polynomial that divides all of the $f_1,\dots,f_s$ also divides h (it is the greatest of all common divisors).

Notation: $h = \gcd(f_1,\dots,f_s)$.

Next, we give a proposition that explains how to calculate $\gcd(f_1,\dots,f_s)$ for $s \geq 2$ iteratively using the base case $s=2$, which we have already seen and which gives the interpretation of $\gcd(f_1,\dots,f_s)$ in terms of ideals. The proposition also clarifies uniqueness. So, we have all of the properties of the claims of Proposition 1.5.7 extended to the case of $s \geq 2$ polynomials.

1.5.9. Proposition For $s \geq 2$ polynomials $f_1, \dots, f_s \in k[x]$, the following assertions hold:

(i) $\gcd(f_1, \dots, f_s)$ is unique up to a nonzero multiple from $k \setminus \{0\}$.

(ii) $\langle \gcd(f_1, \dots, f_s) \rangle = \langle f_1, \dots, f_s \rangle$ (that is, $\gcd(f_1, \dots, f_s)$ is characterized as the generator of the ideal $\langle f_1, \dots, f_s \rangle$).

(iii) $\gcd(f_1, \dots, f_s) = \gcd(f_1, \gcd(f_2, \dots, f_s))$.

(iv) There exists an algorithm that calculates $\gcd(f_1, \dots, f_s)$ for given $f_1, \dots, f_s \in k[x]$.

Proof: To show (i) and (ii) just repeat the arguments in the proof of assertions (i) and (ii) of Proposition 1.5.7. Those assertions were for two polynomials. Now, we have s polynomials but the arguments are completely the same.

(iii) can be verified either directly or as a consequence of (ii). Verifying using (ii) is probably the more interesting way to do it.

We know $\langle \gcd(f_2, \dots, f_s) \rangle = \langle f_2, \dots, f_s \rangle$

So,

$$\langle f_1, \dots, f_s \rangle =$$

$$= \{ f_1 h_1 + f_2 h_2 + \dots + f_s h_s : h_1, \dots, h_s \in k[x] \}$$

($f_2 h_2 + \dots + f_s h_s$ can be replaced by a multiple of $\gcd(f_2, \dots, f_s)$ in view of this identity)

$= \langle f_1, \gcd(f_2, \dots, f_s) \rangle = \langle \gcd(f_1, \gcd(f_2, \dots, f_s)) \rangle$

here again we use the description of the gcd in terms of ideals given in (i)

$\Rightarrow$

$\langle f_1, \dots, f_s \rangle = \langle \underbrace{\gcd(f_1, \gcd(f_2, \dots, f_s))} \rangle$

this is $\gcd(f_1, \dots, f_s)$ by the description of gcd in terms of ideals, given in (i).

In view of (iii), we can verify (iv) by showing that calculation of the gcd of s polynomials reduces to the iterative process, in which the gcd of two polynomials is calculated in each iteration

GCD$(f_1, \dots, f_s)$

Assumption: $f_1, \dots, f_s \in K[x]$

Result: $\gcd(f_1, \dots, f_s)$.

```
1:  i := s
2:  g := f_s        ▷ this is the gcd in progress
3:  ▷ Invariant of the loop: g = gcd(f_i, ..., f_s).
4:  while i > 1 :   ▷ work to be done?
5:      g := greatest_common_divisor(f_{i-1}, g)
            ▷ now f_{i-1} is also accounted for
6:      i := i - 1  ▷ update i to maintain the
                       invariant in the next
                       iteration
7:  end while
8:  return g        ▷ as i = 1, this is the answer.
```

There is an alternative approach to solve this algorithmic problem. We could have extended the Euclidean Algorithm from two polynomials to $s \geq 2$ polynomials. □

With these results we are ready to solve the ideal membership problem for the ring $k[x]$:

deciding if a given polynomial belongs to a given ideal $\langle f_1, \dots, f_s \rangle$ when $f_1, \dots, f_s \in k[x]$.

1.5.10 Corollary There is an algorithm that decides if a polynomial f is in the ideal $\langle f_1, \dots, f_s \rangle$ for given f and $f_1, \dots, f_s$ from $k[x]$.

Proof: Calculate $\gcd(f_1, \dots, f_s)$ and check if f is divisible by $\gcd(f_1, \dots, f_s)$.

Since $\langle f_1, \dots, f_s \rangle = \langle \gcd(f_1, \dots, f_s) \rangle$ (by Proposition 1.5.9 (iii)) and since $\gcd(f_1, \dots, f_s)$ can be computed (by Proposition 1.5.9 (iv)), this procedure solves the decision problem $f \in \langle f_1, \dots, f_s \rangle$. □

1.5.11 Exercises This section had algorithms so that we can have computational exercises just to try the algorithms out. Otherwise, we can see how these algorithms

help solving systems of polynomial equations in one variable when the underlying coefficient field is algebraically closed. Concretely, we consider the field $k = \mathbb{C}$ of complex numbers.

(1) Carry out long division of $x^3 + 2x^2 + x + 1$ by $2x + 1$ in $\mathbb{Q}[x]$.

(2) Determine $\gcd(x^4 - 1, x^6 - 1)$ in $\mathbb{Q}[x]$

(3) Show that every polynomial $f \in \mathbb{C}[x]$ of degree $n \geq 1$ can be factorized as $f = c(x - r_1)^{m_1} \cdot \ldots \cdot (x - r_e)^{m_e}$ where $c \in \mathbb{C} \setminus \{0\}$, $r_1, \ldots, r_e$ are distinct roots of f, $m_1, \ldots, m_e \geq 0$ and $m_1 + \ldots + m_e = n$. Use the fact that $\mathbb{C}$ is algebraically closed.

(4) This one is a follow-up on the Euclidean Algorithm. Show that for $f, g \in k[x]$ one has

$\gcd(f, g) = Af + Bg$

for some $A, B \in k[x]$.

Keywords: Extended Euclidean Algorithm (for polynomials)

this equality is called the Bézout identity.

(5) Is $x^3 + 4x^2 + 3x + 7 \in \langle x^3 - 3x + 2, x^4 - 1, x^6 - 1 \rangle$?

The ring here is $\mathbb{Q}[x]$.

⑥ Consider the system of polynomial equations $f_1 = \dots = f_s = 0$ in one variable x and with coefficients in $\mathbb{C}$.
Design an algorithm that verifies if this system has a solution in $\mathbb{C}$.
The arithmetic of $\mathbb{C}$ is assumed to be available.

⑦ You must have had some experience with calculus in your Bachelor program. So, I guess you know what a derivative is. While one cannot do derivatives over any field the same way as one does them for the fields $k=\mathbb{R}$ and $k=\mathbb{C}$ in calculus, one can define the derivative f' of a polynomial $f \in k[x]$ for any coefficient field k completely formally:

If $f = \sum_{j=0}^{\infty} c_j x^j$, then f' is defined by $f' = \sum_{j=1}^{\infty} j \cdot c_j x^{j-1}$.

Show $(f+g)' = f' + g'$

$(f \cdot g)' = f'g + fg'$

⑧ Assume that a polynomial $f \in \mathbb{C}[x]$ of degree ≥ 1 has been factorized as in ③:

$$f = c(x-r_1)^{m_1} \cdot \dots \cdot (x-r_\ell)^{m_\ell}$$

Find a respective factorization of $\gcd(f, f')$ and $\frac{f}{\gcd(f, f')}$.

(9) If you have solved the above exercises, you probably understand that the system $f_1 = \dots = f_s = 0$ with $f_1, \dots, f_s \in k[x]$ gets reduced to just one equation $f = 0$. Its solution set is the variety $V(f)$.

Show for $k = \mathbb{C}$, that $I(V(f)) = \left\langle \frac{f}{\gcd(f, f')} \right\rangle$.

2 Gröbner bases

Let's move from ideals in $k[x]$ to ideals in $k[x_1, \dots, x_n]$. We are interested in qualitative and algorithmic theory of such ideals. Our intention is to use this understanding for solving various natural problems from algebra and geometry (related to the varieties in k^n). We are inspired by the case $n=1$, we have analyzed above, but the general case $n \in \mathbb{N}$ is more involved (and more interesting). After all algebra of $k[x_1, \dots, x_n]$ is richer than algebra of $k[x]$ and, respectively, the geometry in k^n is richer than the geometry in k^1.

2.1 Monomial orders

What do we need to do to carry over the long division and gcd algorithms to the multivariate setting ? In the case of one variable, our monomials were naturally ordered by degree. The leading term was defined based on this order and long division doesn't

need anything but the access to the leading term function and the operations $+$ and $\cdot$ on $k[x]$.

Say, how to order x^2y, xy^2 and xyz?
The answer is: there is no unique way to do that. In the multivariate case, there are many ways to order monomials.

Still, we need to define what a reasonable ordering is. Let's first clarify what it means to order a set.

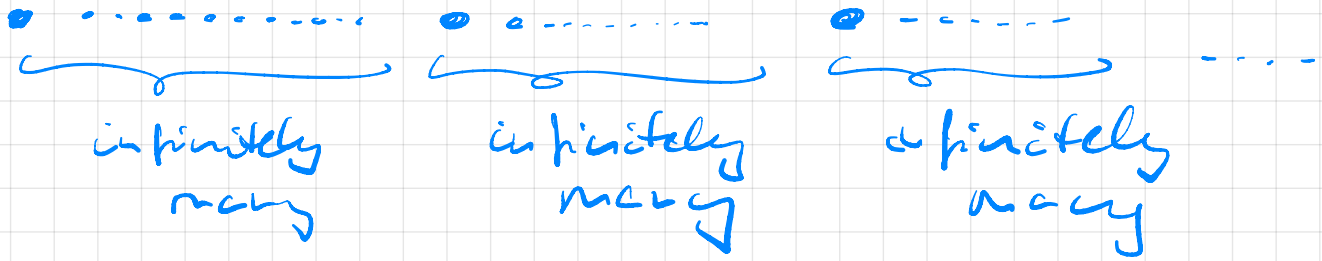
2.1 Definition Consider a binary relation $\preceq$ (usually $\leq$) on a set M. We say that $\preceq$ is a non-strict total order on M if the following hold:

Antisymmetry: $x \preceq y, y \preceq x \Rightarrow x = y$
for all $x, y \in M$

Transitivity: $x \preceq y, y \preceq z \Rightarrow x \preceq z$
for all $x, y, z \in M$

Totality: one has $x \preceq y$ or $y \preceq x$
for all $x, y \in M$.

For a finite M it is like listing the elements of M in some order in an array. But for an infinite M, such an ordering is not necessarily resembling an infinite array. E.g.:



The above picture sketches a situation where a linear ordering is an infinite concatenation of infinite arrays.

Having a nonstrict total order $\succsim$ on M, we can also consider the respective strict total order $\succ$ (cmp. 2) given by $x \succ y \iff x \succsim y, x \neq y$

Clearly, $\succsim$ determines $\succ$ and, vice versa, $\succ$ determines $\succsim$. So, it's enough to describe one of them when we totally order some given set M.

We want to order the set of all monomials in $x = (x_1, \dots, x_n)$:

$\{ x^\alpha : \alpha \in \mathbb{Z}_{\geq 0}^n \}$. Monomials are determined by their exponent vectors. This means, we want to totally order the set $\mathbb{Z}_{\geq 0}^n$ of all exponent vectors.

For $n=1$, our ordering was:

$$x^0 \succ x^1 \succ x^2 \succ x^3 \succ \dots$$

corresponding to $0 < 1 < 2 < 3 < \dots$

The leading term $LT(f)$ of a univariate polynomial $f \in k[x] \setminus \{0\}$ is the term of the largest monomial with respect to the above order.

The long division algorithm in the univariate case multiplies polynomials with terms.

Therefore, it relies on the following property:

$$x^\alpha > x^\beta \Rightarrow x^\alpha x^\gamma > x^\beta x^\gamma \quad (\alpha, \beta, \gamma \in \mathbb{Z}_{\geq 0}^n)$$

That means, the order is preserved under multiplication with any monomial.

There is yet another crucial property: When we carry out the long division the degree of the remainder r in progress gets smaller in each iteration and the termination criterion $\deg r < \deg g$ is achieved after finitely many iterations. For the monomial orders in the multivariate case, we need a condition that would guarantee finite termination. The above discussion provides motivation for the following definition.

2.1.2. Definition A strict monomial order $>$ on $\mathbb{Z}_{\geq 0}^n$ is a binary relation on $\mathbb{Z}_{\geq 0}^n$ such that:

(i) $>$ is a strict total order on $\mathbb{Z}_{\geq 0}^n$

(ii) If $\alpha, \beta, \gamma \in \mathbb{Z}_{\geq 0}^n$, then $\alpha > \beta$ implies $\alpha + \gamma > \beta + \gamma$.

(iii) Every nonempty subset A of $\mathbb{Z}_{\geq 0}^n$ has the

minimal element w.r.t. $\leq$, which is an element $\beta \in A$ satisfying $\alpha \not\leq \beta$ for all $\alpha \in A$, $\alpha \neq \beta$.

By identifying $\mathbb{Z}_{\geq 0}^n$ with $\{x^\alpha : \alpha \in \mathbb{Z}_{\geq 0}^n\}$, we also speak about monomial orders on $\{x^\alpha : \alpha \in \mathbb{Z}_{\geq 0}^n\}$.

Remarks

(i) guarantees that any two monomials are comparable.

(ii) guarantees compatibility with the multiplication by a monomial.

(iii) guarantees that the multivariate long division (to be introduced) terminates. Only infinite sets A matter in (iii), as finite ones have the minimum w.r.t. any linear order.

Rem A set with a total order is called totally ordered (surprise :)). It's up to me to choose if you want to define the strict or the non strict one.

2.1.3 Definition A descending chain of a totally ordered set $(M, \geq)$ is a sequence of elements that satisfy $a_0 \geq a_1 \geq a_2 \geq \dots$.
A chain can be finite or countably infinite.

2.1.4 Lemma Let $(M, \geq)$ be a totally ordered set. Then the following are equivalent:

(i) Every descending chain of M is finite

(ii) Every nonempty subset A of M has the minimal element w.r.t. the order $\geq$.

Proof: It's enough to verify the contraposition $\overline{(i)} \Leftrightarrow \overline{(ii)}$.

If some nonempty set $A \subseteq M$ has no minimal element w.r.t. $\geq$, we can pick any $a_0 \in A$. Since a_0 is not minimal in A, there exists $a_1 \in A$ with $a_0 \geq a_1$. And so on to infinity, we get the infinite chain $a_0 \geq a_1 \geq a_2 \geq \dots$ of elements.
Conversely, if $a_0 \geq a_1 \geq a_2 \geq \dots$ is an infinite chain, then the set $\{a_i : i \in \mathbb{N}_0\}$ has no minimal element because for every a_i, the element a_{i+1} is even smaller. □

There are infinitely many choices of different monomial orders. We probably don't have time to consider all of them :) So, let's stick to the following three that turn out to be rather useful.

2.1.5. Definition We introduce orders:

lex = lexicographic order

grlex = graded lexicographic order (deglex is another common abbreviation)

grevlex = graded reverse lexicographic order

For $\alpha = (\alpha_1, \dots, \alpha_n), \beta = (\beta_1, \dots, \beta_n) \in \mathbb{Z}_{\geq 0}^n$

$\alpha >_{lex} \beta :\Leftrightarrow \alpha_1 = \beta_1, \dots, \alpha_{i-1} = \beta_{i-1}, \alpha_i > \beta_i$ for some $i = 1, \dots, n$ (dictionary order: in our dictionary all words have length n and the alphabet is $\mathbb{Z}_{\geq 0}$).

$\alpha >_{grlex} \beta :\Leftrightarrow |\alpha| > |\beta|$ or $|\alpha| = |\beta|, \alpha >_{lex} \beta$ (order by total degree, if equal tie break by lex).

$\alpha >_{grevlex} \beta :\Leftrightarrow |\alpha| > |\beta|$ or $\alpha_i < \beta_i, \alpha_{i+1} = \beta_{i+1}, \dots, \alpha_n = \beta_n$ for some $i = 1 \dots n$

(order by total degree, if equal unusual tie break).

Remarks lex and grlex look rather natural (right?). lex can be used for elimination. grlex has a better control of the number of iterations in the long division. grevlex is even better in this respect.